

Towards reinforcement learning in UAV relay for anti-jamming maritime communications

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ABSTRACT

Maritime communications with sea surface reflections and sea wave occlusions are susceptible to jamming attacks due to the wide geographical area and intensive wireless communication services. Unmanned Aerial Vehicles (UAVs) help relay messages to improve communication performance, but the relay policy that depends on the rapidly changing maritime environments is difficult to optimize. In this paper, a reinforcement learning-based UAV relay policy for maritime communications is proposed to resist jamming attacks. Based on previous transmission performance, the relay location, the received power of the transmitted signal and the received jamming power, this scheme optimizes the UAV trajectory and relay power to save the energy consumption and decrease the Bit-Error-Rate (BER) of the maritime signals. A deep reinforcement learning-based scheme is also proposed, which designs a deep neural network with dueling architecture to further improve the communication performance and computational complexity. The performance bounds regarding the signal to interference plus noise ratio, energy consumption and the communication utility are provided based on the Nash equilibrium of the game against jamming, and the computational complexity of the proposed schemes is analyzed. Simulation results show that the proposed schemes improve the energy efficiency and decrease the BER compared with the benchmark.

1. Introduction

Anti-jamming maritime communications are critical to support environment monitoring, naval surveillance, maritime rescue, and mineral exploration [1,2]. Due to the sea surface motions, wave occlusions, and harsh weather conditions such as storms, maritime communications are challenged by complicated channel conditions, severe channel fading, and strong jamming. The power control and channel allocation that determines energy consumption and transmission throughput have been optimized by convex optimization [3,4] based on channel gain against jamming attacks, which are rarely known by practical maritime communication systems.

To improve the anti-jamming performance under the unknown channel models and jamming models, Reinforcement Learning (RL) algorithms such as Q-learning and Deep Q-Network (DQN) have been used to optimize the anti-jamming policy. For example, the Q-learning based multiple-relay cooperative network in [5] improves the power allocation of both transmitters and relay nodes without being aware of the

instantaneous channel state information. The game theory study on the anti-jamming interactions [6,7] provides the existence conditions of the Nash Equilibrium (NE) and Stackelberg Equilibrium (SE) of the games and provides the performance bounds of the RL schemes. However, these schemes suffer from severe performance degradation under maritime environments due to limited base stations and communication devices especially in the open oceans [8].

The anti-jamming maritime communication performance can be improved by exploiting the line-of-sights and spatial diversity of Unmanned Aerial Vehicles (UAVs) [9]. Because of the easy deployment and lower path-loss exponents, UAVs can provide better channel states to avoid the ongoing ship-to-ship communications being blocked by a jammer with strong jamming power. Therefore, the maritime communication systems use the UAV as a relay to forward the maritime message to improve the data rate of the information transmission and reduce the Bit Error Rate (BER) of maritime messages against the jammer that can significantly degrade the received quality of the radio signals and increase the transmission delay.

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In this paper, a UAV-aided framework for ship-to-ship communications is proposed against jamming attacks in dynamic maritime environments without knowing the complete information about jamming models and communication channels. This framework utilizes the UAV to relay maritime messages for ships to resist the jammers and designs two RL algorithms to optimize the relay policy. The UAV chooses the relay power, the moving distance and the moving direction in each time slot according to Q-values based on the current state that consists of the previous BER from the destination ship, the received power from the source ship, the received jamming power, and the previous location of the UAV. The UAV utility consists of the Signal to Interference plus Noise Ratio (SINR) of the maritime signal and the energy consumption of the UAV mobility and transmission.

A Double Q-learning-based Relay (DQR) scheme is proposed to choose the relay power, the moving distance and direction for the relay UAV based on the expected long-term discounted reward for state-action pairs in each time slot. The double Q-learning [10] can obtain the optimal relay policy via the ϵ -greedy algorithm to maximize Q-values that are updated by the Bellman function. Moreover, the double Q-learning of the proposed DQR scheme applies two Q-tables with different subsets of experience samples to evaluate the UAV policy, which solves the over-estimation dilemma of the single Q-table. In order to further improve the learning efficiency and communication performance, a Dueling DQN-based Relay (DDQNR) scheme that designs a Deep Neural Network (DNN) with dueling architecture is proposed to solve the high-dimensional states. Different from [11], the DNN in the DDQNR scheme using the Fully Connected (FC) layer instead of three convolutional layers can alleviate the computational complexity of the anti-jamming communication algorithm. The network followed by the FC layer is separated into two FC layers as two estimators for the state-value and state-dependent action advantages, respectively.

The computational complexity of the proposed schemes that depends on the size of action space, the batch size and the number of neural nodes is analyzed. The interactions between the UAV and the jammer is modeled as a relay game and the performance bounds regarding the SINR, energy consumption and utility are analyzed. The NEs of the game is derived and the impact of the maritime channel states and cost coefficients on the BER of the maritime messages and energy consumption of the UAV is analyzed. Simulation results based on a UAV-aided system show that the proposed schemes achieve higher utility, lower BER and energy consumption compared with the benchmark.

The main contributions are summarized as follows:

- A DQR UAV relay scheme is proposed to jointly optimize the relay power and moving policy of the relay UAV for ship-to-ship communications against jamming attacks in dynamic maritime environments without knowing the channel parameters and jamming model.
- A DDQNR scheme is proposed to accelerate learning for the relay UAV with numerous quantifications of states and improve the anti-jamming maritime communication efficiency.
- The performance bounds via the relay game, the conditions under which the NEs exist, and the computational complexity of the proposed schemes are provided.

Section 2 reviews related work, and Section 3 presents the UAV-assisted maritime anti-jamming communication model. The DQR scheme is proposed in Section 4 and the DDQNR scheme is proposed in Section 5. Performance analysis and simulation results are presented in Sections 6 and 7, respectively. Section 8 concludes the paper.

2. Related work

To improve the communication performance in wireless networks, anti-jamming has been widely studied in [12,13]. For example, the scheme proposed in [12] applies a frequency-hopping technique to enhance the anti-jamming capability and provide a robust tactical data

link for a UAV relay under a jamming attack. An anti-jamming scheme in [14] dynamically selects both the frequency hopping and transmission rate to obtain a better throughput and jamming resiliency to resist the reactive-sweep jammer in wireless communications. A cooperative anti-jamming scheme as proposed in [15] jointly optimizes the channel access probability at the medium access control layer and relays node selection at the physical layer with a pricing-based distributed algorithm to enhance network traffics under different jamming strategies. A sleeping scheduling and opportunistic transmission scheme in [16] employs a delay-tolerant node to relay vessel communications in maritime wireless networks, which uses the Lyapunov optimization and convex optimization methods to derive the best tradeoff between the delay and energy consumption under the assumption that the system could detect the channel status.

UAVs have drawn increasing attention in improving the communication performance for maritime communications due to their versatility. The UAV is applied as relay stations to enhance the communication coverage of communication networks [17], collect data to support maritime activities [18] and provide wireless connectivity for devices [19]. In addition, the UAVs are also deployed in the mesh network [20] and the cellular network [21] to support high-performance maritime communications. For example, the UAV-aided scheme in [22] uses a UAV to relay mobile user messages to a base station to improve the communication efficiency for the anti-jamming cellular communication system. For the sake of improving the energy efficiency of the relay UAV, energy harvesting [23], beamforming [24] and trajectory optimizing [25] have been developed. For instance, a UAV-aided maritime radar surveillance system in [25] uses a multiobjective particle swarm optimization algorithm to optimize global UAV trajectories for energy saving. A joint link scheduling and rate adaptation scheme is proposed in [26] to minimize the energy consumption of the satellite-UAV-terrestrial networks with the quality of service guarantees for maritime coverage enhancement, which uses only the large-scale channel state information.

RL-based anti-jamming maritime communications optimize the anti-jamming policy such as power control and frequency hopping without relying on the known jamming and channel models for the vehicular ad hoc networks [27], cellular networks [28], underwater sensor networks [29], Internet of satellites [30] and intelligent reflecting surface-assisted system [31]. In particular, the UAV applies a policy hill-climbing algorithm to decide whether to relay messages with fixed relay power and position for vehicular ad hoc networks [27], which reduces the BER of vehicle messages. The hotbooting Q-learning and hotbooting deep Q-networks algorithms are developed in [29] to improve the BER performance and reduce the energy consumption of relay sensors against jamming for underwater sensor networks. A spatial anti-jamming scheme in [30] proposes a deep RL algorithm to obtain an available routing subset for heterogeneous Internet of Satellites. A fast-policy hill-climbing learning approach is proposed in [31] to achieve the optimal power allocation and reflection beamforming policy to improve the data rate and SINR for intelligent reflecting surface-assisted wireless communications against jamming attacks.

Game theory theoretic [6,7] study on interactions between the defender and the jammer analyzes the existence conditions of the equilibrium of the games to disclose the performance bounds of the anti-jamming communication schemes. For instance, game theory is applied in [6] to investigate the channel selection and jamming strategy for mitigating interference in dense wireless networks, and the Nash equilibria solution is derived based on the stochastic learning theory. The distributed anti-jamming scheme in [7] applies the Stackelberg game to formulate the interaction between multiple smart jammers and satellite-enabled army IoT devices, and then proposes a modified coalition formation game to reduce the energy consumption for devices.

3. System model

In this system, a maritime communication network consisting of two

ships, a drone, and a jammer is constructed, as shown in Fig. 1. Ship A communicates with ship B to share maritime information such as navigation and distress alerts. It is assumed that the A-B ship link is affected by a smart jammer that can intelligently attack legitimate communications between A and B ships. In order to reduce the bit error rate of the signal received by ship B, the UAV is used as a relay to resist maritime communication interference.

3.1. Network model

Ship A transmits messages to ship B at the frequency f_c with the bandwidth W and the transmission power P_T . In order to resist the smart jamming attacks, a UAV is deployed to relay the transmitted messages for better communication performance.

It assumes that the relay UAV with the fixed flight altitude H . At time slot k , the relay UAV receives the BER from ship B via the feedback channel, meanwhile estimates the received jamming power $\hat{p}_j^{(k)}$ and the received transmit power $\hat{p}_T^{(k)}$. After that, the relay UAV selects the moving distance $l^{(k)} \in \{gD/N | 0 \leq g \leq N\}$ and the moving direction $\theta^{(k)} \in \{2\pi z/Z | 0 \leq z \leq Z\}$ to move from the location $\mathbf{L}^{(k-1)}$ to the next location $\mathbf{L}^{(k)}$, where D is the maximum moving distance with N levels to ensure the remote control requirements for UAV, Z is the quantization levels of the moving direction. And then, the UAV relays the maritime messages to ship B with power $p_R^{(k)} \in \{jP_R/X | 0 \leq j \leq X\}$, and P_R is the maximum transmit power of the relay UAV with X levels.

Upon the received signals from ship A and the relay UAV, ship B decodes the maritime messages and combines the two signals via the selection combining method [32]. Ship B evaluates the BER of the received messages $b^{(k)}$ and then returns to the relay UAV through the feedback channel. So as to satisfy the communication requirements in maritime environments, the BER should be smaller than a predefined threshold to avoid communication outages.

The jammer aims to block the legitimate links of the ship-ship communication and the ship-UAV communication by transmitting jamming signals. The smart jammer estimates the transmit power of ship A and the UAV. In addition, the smart jammer applies smart radio devices to eavesdrop on the control channel of the UAV and estimate the UAV transmission policy [29]. In order to degrade even break the legitimate maritime communication, the smart jammer applies the RL technique to choose the jamming power $p_j^{(k)} \in \{tP_J/G | 0 \leq t \leq G\}$ and sends the faked or replayed signals at the center frequency f_c and the bandwidth W to ship B, and P_J is the maximum jamming power with G levels.

3.2. Channel model

Similar to literature [3], the path loss of the ship-UAV and

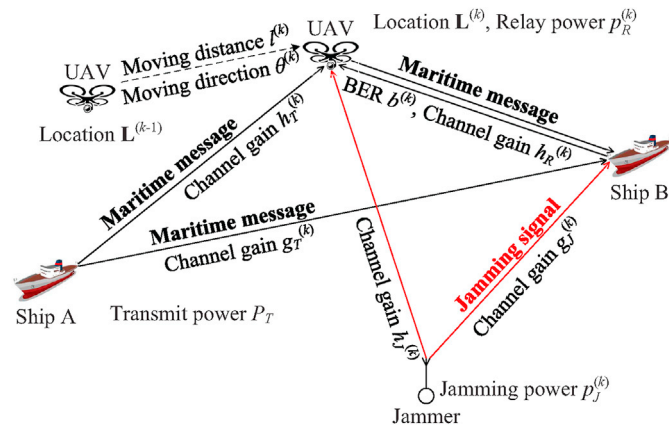


Fig. 1. System model: ship A sends static and voyage related information to ship B with transmit power $P_T^{(k)}$ against a smart jammer with power $p_j^{(k)}$.

jammer-UAV links at distance d is calculated by

$$PL = PL_0 + 10\eta \log\left(\frac{d}{d_0}\right) + w_0, d \geq d_0 \quad (1)$$

where d_0 is the reference distance, PL_0 is the path loss in dB at d_0 , η is the path-loss exponent, and w_0 is a Gaussian random variable with zero-mean and deviation δ .

The maritime communication suffers from severe multipath effects due to the sea surface reflection, the path loss of ship A-B and jammer-ship B at distance d can be calculated by the 2-ray path loss model in [9] as

$$PL_{2\text{-ray}} = -20\log\left(\frac{\lambda}{2\pi d} \sin\left(\frac{2\pi H_A H_B}{\lambda d}\right)\right) \quad (2)$$

where λ is the wavelength of the transmitted signal, and the antenna heights of ship A and B are denoted as H_A and H_B , respectively.

At time slot k , the channel gain vector $\mathbf{h}^{(k)}$ includes the channel gains of ship A-UAV $h_T^{(k)}$, ship A-B $g_T^{(k)}$, jammer-ship B $g_j^{(k)}$, jammer-UAV $h_j^{(k)}$ and UAV-ship B $h_R^{(k)}$. At time slot k , under the noise signal n_0 with power σ , the SINR of ship A-B link at the ship B can be expressed as $\zeta^{(k)}$, the SINR of the first-hop ship-ship link at UAV can be expressed as $\zeta^{(k)}$, the SINR at ship B in the second-hop ship-ship link can be given by $\varphi^{(k)}$. Similar to [33], the overall SINR of the dual-hop amplify-and-forward link can be calculated as

$$\beta^{(k)} = \frac{\varphi^{(k)} \varphi^{(k)}}{\varphi^{(k)} + \varphi^{(k)} + 1} \quad (3)$$

and the BER with Quadrature Phase Shift Keying (QPSK) can be calculated as

$$b^{(k)} = 0.5 \operatorname{erfc}\left(\sqrt{0.5 \max(\beta^{(k)}, \zeta^{(k)})}\right) \quad (4)$$

The key notations are listed in Table 1 for reference.

4. RL-based UAV relay scheme against jamming attacks

In the maritime communication environment, the UAV can hardly obtain the channel status information and the jamming attack model. Therefore, a DQR scheme that applies a double Q-learning algorithm to optimize the UAV relay policy is proposed to resist jamming attacks for maritime communications, avoiding the overestimation dilemma of single Q-table. In particular, the relay UAV chooses the moving distance, moving direction and relay power based on the state consisting of the BER of the received signals at the last time slot by ships, the received power of the transmitted signal, the received jamming power and the UAV location at last time slot.

Table 1
Summary of key symbols and notations.

Notation	Definition
P_T	Transmit power of ship A
$p_R^{(k)} \in \{jP_R/X 0 \leq j \leq X\}$	Relay power of UAV
$p_j^{(k)} \in \{tP_J/G 0 \leq t \leq G\}$	Jamming power
$\mathbf{h}^{(k)}$	Channel gain vector
$\hat{p}_T^{(k)}/\hat{p}_j^{(k)}$	Received power of the transmit/jamming signal
$\theta^{(k)} \in \{2\pi z/Z 0 \leq z \leq Z\}$	Moving direction of UAV
$l^{(k)} \in \{gD/N 0 \leq g \leq N\}$	Moving distance of UAV
$\mathbf{L}^{(k)}$	UAV Location
$b^{(k)}$	BER of the messages received by ship B
H_A/H_B	Antenna height of ship A/B
d_0	Reference distance
PL_0	Path loss at d_0
η	Path loss exponent

The relay UAV obtains the previous BER $b^{(k-1)}$ of the received message from ship B and the received signal power $\hat{p}_T^{(k)}$, measures the received jamming power $\hat{p}_J^{(k)}$ and observes the previous UAV location $\mathbf{L}^{(k-1)}$ at time slot k to formulate the state $\mathbf{s}^{(k)}$, i.e.

$$\mathbf{s}^{(k)} = [b^{(k-1)}, \hat{p}_T^{(k)}, \hat{p}_J^{(k)}, \mathbf{L}^{(k-1)}] \quad (5)$$

Algorithm 1: The DQR scheme for maritime communications

```

1: Initialize  $\alpha, \gamma, Q_1(s, \mathbf{x}), Q_2(s, \mathbf{x}), \mathbf{L}^{(0)}$ 
2: for  $k = 1, 2, 3, \dots$ , do
3:   Measure  $\hat{p}_T^{(k)}$  and  $\hat{p}_J^{(k)}$ 
4:    $\mathbf{s}^{(k)} = [b^{(k-1)}, \hat{p}_T^{(k)}, \hat{p}_J^{(k)}, \mathbf{L}^{(k-1)}]$ 
5:   Choose  $\mathbf{x}^{(k)}$  via  $\epsilon$ -greedy
6:   Move to a new location  $\mathbf{L}^{(k)}$ 
7:   Relay maritime messages with power  $p_R^{(k)}$ 
8:   Receive  $b^{(k)}$  from the feedback channel
9:   Calculate  $u^{(k)}$  via (6)
10:  Sample  $q$  uniformly,  $q \sim U[0, 1]$ 
11:  if  $q < 0.5$  then
12:     $\mathbf{x}^* = \arg \max_{\mathbf{x}'} Q_1(s^{(k+1)}, \mathbf{x}')$ 
13:     $Q_1(s^{(k)}, \mathbf{x}^{(k)}) \leftarrow (1 - \alpha)Q_1(s^{(k)}, \mathbf{x}^{(k)})$ 
14:       $+ \alpha(u^{(k)} + \gamma Q_2(s^{(k+1)}, \mathbf{x}^*))$ 
15:  else if  $q \geq 0.5$  then
16:     $\mathbf{x}^* = \arg \max_{\mathbf{x}'} Q_2(s^{(k+1)}, \mathbf{x}')$ 
17:     $Q_2(s^{(k)}, \mathbf{x}^{(k)}) \leftarrow (1 - \alpha)Q_2(s^{(k)}, \mathbf{x}^{(k)})$ 
18:       $+ \alpha(u^{(k)} + \gamma Q_1(s^{(k+1)}, \mathbf{x}^*))$ 
19:  end if
20: end for

```

The UAV is assumed to arrive at the location $\mathbf{L}^{(k)}$ with distance $l^{(k)}$ and direction $\theta^{(k)}$, and then relays the signal with power $p_R^{(k)}$. The energy consumption of the relay UAV includes the relay power consumption and the moving energy consumption, and the utility $u^{(k)}$ of UAV is calculated by

$$u^{(k)} = \text{SINR}^{(k)} - c_1 p_R^{(k)} - c_2 l^{(k)} \quad (6)$$

where the SINR in dB is calculated by $\text{SINR}^{(k)} = 20 \log \text{erfcinv}(2b^{(k)}) + 3.01$, c_1 is the cost coefficient of the relay power, c_2 is the unit cost of the moving distance and $\text{erfcinv}(\cdot)$ is the inverse function of $\text{erfc}(\cdot)$. In order to increase the relay utility and decrease the BER and the energy consumption, the DQR scheme adopts a double-Q learning algorithm with two Q-tables Q_1 and Q_2 to optimize the UAV relay policy.

The UAV has $X + 1$ transmit power levels, $Z + 1$ moving direction levels and $N + 1$ moving distance levels, and the dimensional sequence of action space is denoted as $|\mathbf{A}| = (N + 1)(X + 1)(Z + 1)$. At time slot k , the UAV obtains information and formulates state $\mathbf{s}^{(k)}$, then selects action $\mathbf{x}^{(k)} = [p_R^{(k)}, l^{(k)}, \theta^{(k)}]$ based on Q_1 and Q_2 with a high probability of $1 - \epsilon$ to maximize the current Q-values and other policies with probability ϵ . After that, the UAV performs the moving policy with $\mathbf{L}^{(k)} = \mathbf{L}^{(k-1)} + [l^{(k)} \cos \theta^{(k)}, l^{(k)} \sin \theta^{(k)}, 0]$ and relays the maritime messages to ship B with power $p_R^{(k)}$. Each Q-function is updated with a value from the other Q-function for the next state. The Q-values are updated by the Bellman function, which is given by

$$Q(s^{(k)}, \mathbf{x}^{(k)}) \leftarrow (1 - \alpha)Q(s^{(k)}, \mathbf{x}^{(k)}) + \alpha(u^{(k)} + \gamma \max_{\mathbf{x}'} Q(s^{(k+1)}, \mathbf{x}')) \quad (7)$$

The overall description of the DQR scheme is shown in Algorithm 1.

5. Deep RL-based UAV relay scheme against jamming attacks

Algorithm 2: The DDQNR scheme for maritime communications

```

1: Initialize  $\beta, \gamma, \mathbf{L}^{(0)}, \varphi^{(1)}, \theta_2, \theta_2, T$ 
2: for  $k = 1, 2, 3, \dots$ , do
3:   Measure  $\hat{p}_T^{(k)}$  and  $\hat{p}_J^{(k)}$ 
4:    $\mathbf{s}^{(k)} = [b^{(k-1)}, \hat{p}_T^{(k)}, \hat{p}_J^{(k)}, \mathbf{L}^{(k-1)}]$ 
5:   Input  $\mathbf{s}^{(k)}$  to the DNN
6:   Evaluate the state-value  $V(s^{(k)}; \theta_1^{(k)})$  and the
     action advantage value  $A(s^{(k)}, \mathbf{x}^{(k)}; \theta_2^{(k)})$  with
     the DNN output
7:   Calculate the Q-values via (8)
8:   Choose  $\mathbf{x}^{(k)}$  via  $\epsilon$ -greedy
9:   Move to a new location  $\mathbf{L}^{(k)}$ 
10:  Relay maritime messages with power  $p_R^{(k)}$ 
11:  Receive  $b^{(k)}$  from the feedback channel
12:  Calculate  $u^{(k)}$  via (6)
13:  Formulate  $\mathbf{s}^{(k+1)}$  via (5)
14:   $\mathcal{D} \leftarrow \mathcal{D} \cup \{s^{(k)}, \mathbf{x}^{(k)}, u^{(k)}, s^{(k+1)}\}$ 
15:  for  $j = 1, 2, 3, \dots, T$  do
16:    Select  $\{s^{(j)}, \mathbf{x}^{(j)}, u^{(j)}, s^{(j+1)}\} \in \mathcal{D}$ 
17:  end for
18:  Update  $L(\theta^{(k)})$  via (9)
19:  Update  $\theta^{(k)}$  via (10)
20: end for

```

In this section, a DDQNR scheme for maritime communication against jamming attacks is proposed for the relay UAV with more computing power. The relay UAV formulates and stores the state in the memory pool, and then samples a minibatch from the memory pool. The DNN updates network parameters with the minibatch and outputs the Q-values of the relay policies with the input of the current state. The UAV chooses the relay policy with the max Q-value. Moreover, the DDQNR with a dueling architecture [11] to not only improves the anti-jamming performance by distinguishing the valuable state but also improves the learning speed of the optimal relay policy. Different from the dueling DQN [11], to fully exploit the UAV relay experiences, the DNN exploits two FC layers rather than three convolutional layers to quickly extract the relay features and thus reduce the sample complexity.

At time slot k , the relay UAV formulates the state $\mathbf{s}^{(k)}$, then constructs the state, relay policy and utility as a relay experience, i.e., $\{s^{(k)}, \mathbf{x}^{(k)}, u^{(k)}, s^{(k+1)}\}$, and stores the relay experience in a memory pool $\mathcal{D}$. The DNN used in the DDQNR scheme contains an input layer, two hidden layers and two output layers. As shown in Fig. 2, the input layer involves 4 nodes, FC1 and FC2 consist of m nodes, which is activated by the rectified linear unit, and the weights of FC1 and FC2 are denoted as $\theta_1^{(k)}$ and $\theta_2^{(k)}$, respectively. In addition, the first output layer has $|\mathbf{A}|$ nodes and the second output layer has 1 node. Similar to the dueling DQN algorithm, FC1 layer with parameters $\theta_1^{(k)}$ and FC2 layer with parameter $\theta_2^{(k)}$ are used for separate estimates of advantage function and a scalar value, respectively. As described in Algorithm 2, this scheme inputs the current state $\mathbf{s}^{(k)}$ and outputs the estimated value $Q(s^{(k)}, \mathbf{x}^{(k)}; \theta_1^{(k)}, \theta_2^{(k)})$ at time slot k . Note that the dueling architecture is a deep model that combines the value $V(s^{(k)}; \theta_2^{(k)})$ and advantage $A(s^{(k)}, \mathbf{x}^{(k)}; \theta_1^{(k)})$ to produce the state-action value $Q(s^{(k)}, \mathbf{x}^{(k)}; \theta_1^{(k)}, \theta_2^{(k)})$. As a result, the output of DNN can be calculated as

$$Q(s^{(k)}, \mathbf{x}^{(k)}; \theta_1^{(k)}, \theta_2^{(k)}) = V(s^{(k)}; \theta_2^{(k)}) + \left(A(s^{(k)}, \mathbf{x}^{(k)}; \theta_1^{(k)}) - \frac{1}{|\mathbf{A}|} \sum_{\mathbf{x}'} A(s^{(k)}, \mathbf{x}'; \theta_1^{(k)}) \right) \quad (8)$$

An alternative module replaces the max operator with an average in Eq. (8) according to [11]. The minus of the mean value of advantages in

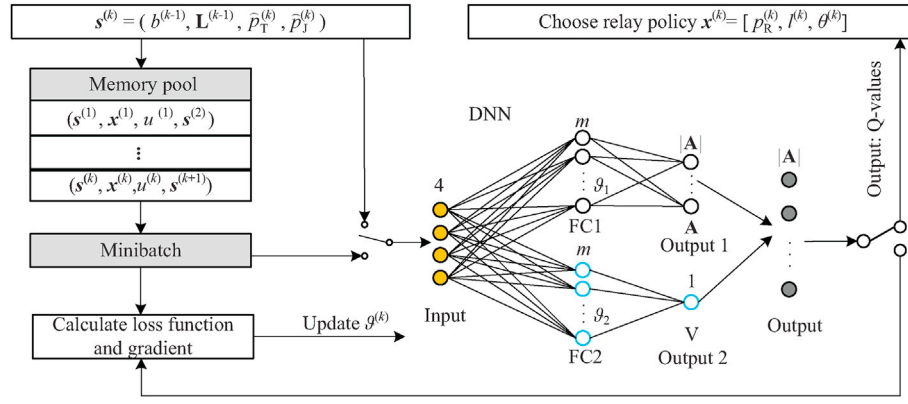


Fig. 2. Dueling DQN based UAV relay for maritime communications.

Eq. (8) does not change the relative rank of the A and Q-values and preserve the greedy or ϵ -greedy policy based on Q-values. The advantages only need to change as fast as the mean with Eq. (8), without compensating any change to the optimal action's advantage. The DNN outputs the action space $|A|$ dimensional sequence, and the relay policies are selected by ϵ -greedy.

The relay UAV selects the relay experience $\{s^{(k)}, x^{(k)}, u^{(k)}, s^{(k+1)}\}$ from $\mathcal{D}$ randomly to update the DNN parameters $\theta^{(k)} = \{\theta_1^{(k)}, \theta_2^{(k)}\}$. The loss function of DNN parameters $\theta^{(k)}$ is written as

$$L(\theta^{(k)}) = \mathbb{E}_{s, x, u, s'} \left[\left(u^{(k)} + \gamma \max_{x'} Q(s', x'; \theta^{(k)}) - Q(s, x; \theta^{(k)}) \right)^2 \right] \quad (9)$$

where s' is the next state sequence. The update of the parameters $\theta^{(k)}$ with respect of the gradient of $L(\theta^{(k)})$ with stochastic gradient descent algorithm is given by

$$\begin{aligned} \theta^{(k+1)} &= \theta^{(k)} + \beta \nabla_{\theta^{(k)}} L(\theta^{(k)}) \\ &= \theta^{(k)} + \beta \left[\mathbb{E}_{s, x, u, s'} \left[\left(u^{(k)} + \gamma \max_{x'} Q(s', x'; \theta^{(k)}) \right) \right. \right. \\ &\quad \times \nabla_{\theta^{(k)}} Q(s, x; \theta^{(k)}) \left. \right] \\ &\quad \left. - \mathbb{E}_{s, x} [Q(s, x; \theta^{(k)}) \nabla_{\theta^{(k)}} Q(s, x; \theta^{(k)})] \right] \end{aligned} \quad (10)$$

where β is the learning rate and γ is the discount factor, and the parameter selection process repeats T times to update $\theta^{(k+1)}$.

6. Performance analysis

In this section, relay performance for maritime communications is analyzed and performance bounds on SINR, energy consumption, and utility are provided. In the maritime anti-jamming communication game, the UAV chooses the relay power $p_R^{(k)} \in [0, P_R]$, the moving distance $l^{(k)} \in [0, D]$ and the moving direction $\theta^{(k)} \in [0, 2\pi]$ to maximize utility $u^{(k)}$, meanwhile, the jammer maximizes utility $u_J^{(k)}$ by choosing the jamming power $p_J^{(k)} \in [0, P_J]$. Specifically, the jamming utility $u_J^{(k)}$ is expressed as

$$u_J^{(k)} = -u^{(k)} - \gamma p_J^{(k)} \quad (11)$$

where γ is the cost coefficient of jamming power. For the sake of making the notations more clear, the time index k is omitted in the following performance analysis.

Two new parameters to indicate the policy of the relay UAV are introduced as d_1 and d_2 . The UAV keeps silent when $d_1 = 0$ while UAV chooses to relay the maritime messages when $d_1 = 1$, and if d_2 is 0, the UAV does not move, if d_2 is 1, the UAV moves to a new location. In this way, the utility of the relay UAV u as $u(d_1, d_2)$ is rewritten. When the UAV

moves to a new location L' , the channel gain vector will be $[h'_T, g_T, g_J, h'_J, h'_R]$ and the SINR at new location is denoted as β' . Four cases of joint decision-making based on whether the UAV is relaying and moving and whether the smart jammer is working are studied, and then the corresponding performance bounds are provided.

Proposition 1. (Case 1) The performance bound of the RL-based relay game under continuous relay policy is given by

$$\begin{cases} \text{SINR} = \frac{g_T P_T}{\sigma} \end{cases} \quad (12a)$$

$$c = 0 \quad (12b)$$

$$u = \frac{g_T P_T}{\sigma} \quad (12c)$$

if the following inequation holds:

$$\min(h_T P_T, h_R P_R) \leq g_T P_T \leq \frac{\gamma \sigma^2}{g_J} \quad (13)$$

Proof. By (11), the first order and second order derivative with respect to p_J are obtained as

$$\frac{\partial u_J}{\partial p_J} = \frac{g_J P_T g_T}{(p_J g_J + \sigma)^2} - \gamma \leq 0 \quad (14)$$

$$\frac{\partial^2 u_J}{\partial p_J^2} = -\frac{2g_J^2 P_T g_T}{(p_J g_J + \sigma)^3} < 0 \quad (15)$$

If (13) holds, it can be inferred the u_J is a monotonic decreasing function of p_J , so u_J maximizes when $p_J = 0$. Thus, $\arg \max_{p_J} u_J = 0$ is proved, and then $g_T P_T \leq \gamma \sigma^2 / g_J$ is derived.

By (6), if (13) holds, $\forall 0 \leq p_R \leq P_R$, it can be inferred that the u is a monotonic decreasing function of p_R due to $\partial u / \partial p_R = -c_1 < 0$ and

$$u(0, 0) = \frac{g_T P_T}{\sigma} \geq \frac{g_T P_T}{\sigma} - c_2 = u(0, 1)$$

Therefore, $\max_x u = u(0, 0)$ is derived. That is to say, the relay UAV achieves maximum utility when the relay UAV does not move and does not relay the maritime messages.

Proposition 2. (Case 2) The performance bound of the RL-based relay game under continuous relay policy is given by

$$\begin{cases} \text{SINR} = \frac{g_T P_T}{\sigma + p_J g_J} \end{cases} \quad (16a)$$

$$c = 0 \quad (16b)$$

$$u = \frac{g_T P_T}{\sigma + p_J g_J} \quad (16c)$$

if the following inequation holds:

$$\min \left(h_T P_T, h_R P_R, \frac{y(p_J g_J + \sigma)^2}{g_J} \right) \leq g_T P_T \quad (17)$$

Proof. By (11), if (17) holds, the first order and second order derivative with respect to p_J are obtained as

$$\frac{\partial u_J}{\partial p_J} = \frac{g_J P_T g_T}{(p_J g_J + \sigma)^2} - y \geq 0 \quad (18)$$

$$\frac{\partial^2 u_J}{\partial p_J^2} = -\frac{2g_J^2 P_T g_T}{(p_J g_J + \sigma)^3} < 0 \quad (19)$$

It is inferred that u_J is maximized when $\partial u_J / \partial p_J = 0$, where $p_J = (\sqrt{g_J P_T g_T} - \sigma) / g_J$. Thus, $\arg \max_J u_J = 1$ is proved.

By (6), if (17) holds, $\forall 0 \leq p_R \leq P_R$, it is noted that the u is a monotonic decreasing function of p_R since $\partial u / \partial p_R = -c_1 < 0$ and

$$u(0, 0) = \frac{g_T P_T}{\sigma + p_J g_J} \geq \frac{g_T P_T}{\sigma + p_J g_J} - c_2 = u(0, 1)$$

Therefore, $\max_{\mathbf{x}} u = u(0, 0)$ is derived. In this case, the relay UAV does not move and relay the maritime messages.

Proposition 3. (Case 3) The performance bound of the RL-based relay game under continuous relay policy is given by

$$\begin{cases} \text{SINR} = \frac{P_T h_T p_R h_R}{\psi} \end{cases} \quad (20a)$$

$$\begin{cases} c = c_1 p_R \end{cases} \quad (20b)$$

$$\begin{cases} u = \frac{P_T h_T p_R h_R}{\psi} - c_1 p_R \end{cases} \quad (20c)$$

if the following inequation holds:

$$c_2 \geq \beta' - \beta \quad (21a)$$

$$h'_T h'_R \leq h_T h_R \quad (21b)$$

$$g_T P_T \leq \min(h_T P_T, h_R P_R) \quad (21c)$$

$$h_T P_T \leq \frac{\sqrt{y P_T h_T p_R h_R (P_T h_T g_J + g_J \sigma)} - y p_R h_R \sigma}{y(p_J g_J + \sigma)} - \sigma \quad (21d)$$

$$p_R \leq \frac{\sqrt{c_1 P_T h_T h_R (\psi' - p_R h_R \sigma)} - c_1 \psi'}{2c_1 h_R \sigma} \quad (21e)$$

where $\psi = P_T h_T p_J g_J + p_J g_J \sigma + P_T h_T \sigma + p_R h_R \sigma + \sigma^2$.

Proof. By (11), if (21) holds, the first order and second order derivative with respect to p_J are obtained as

$$\frac{\partial u_J}{\partial p_J} = \frac{P_T h_T p_R h_R (P_T h_T g_J + g_J \sigma)}{\psi^2} - y \geq 0 \quad (22)$$

$$\frac{\partial^2 u_J}{\partial p_J^2} = -\frac{P_T h_T p_R h_R (P_T h_T g_J + g_J \sigma)}{\psi^3} < 0 \quad (23)$$

It is derived that u_J is maximized when $\partial u_J / \partial p_J = 0$, where $p_J > 0$. Thus, $\arg \max_J u_J = 1$ is proved.

By (6), if (21) holds, $\forall 0 \leq p_R \leq P_R$, it is inferred that the relay UAV obtains the maximum utility when $\partial u_J / \partial p_J = 0$, and

$$\begin{aligned} u(1, 0) &= \frac{P_T h_T p_R h_R}{\psi^2} - c_1 \\ &\geq \frac{P_T h_T p_R h_R}{\psi^2} - c_1 - c_2 = u(1, 1) \end{aligned}$$

Therefore, $\max_{\mathbf{x}} u$ with $\mathbf{x}^* = [1, 0]$ is derived. In this case, the relay UAV relays the maritime messages but not moves.

Proposition 4. (Case 4) The performance bound of the RL-based relay game under continuous relay policy is given by

$$\begin{cases} \text{SINR} = \frac{P_T h'_T p_R h'_R}{\psi'} \end{cases} \quad (24a)$$

$$\begin{cases} c = c_1 p_R - c_2 \end{cases} \quad (24b)$$

$$\begin{cases} u = \frac{P_T h'_T p_R h'_R}{\psi'} - c_1 p_R - c_2 \end{cases} \quad (24c)$$

if the following inequation holds:

$$c_2 < \beta' - \beta \quad (25a)$$

$$h'_T h'_R > h_T h_R \quad (25b)$$

$$g_T P_T \leq \min(h'_T P_T, h'_R P_R) \quad (25c)$$

$$h'_T P_T \leq \frac{\sqrt{y P_T h'_T p_R h'_R (P_T h'_T g_J + g_J \sigma)} - y p_R h'_R \sigma}{y(p_J g_J + \sigma)} - \sigma \quad (25d)$$

$$p_R \leq \frac{\sqrt{c_1 P_T h'_T h'_R (\psi' - p_R h'_R \sigma)} - c_1 \psi'}{2c_1 h'_R \sigma} \quad (25e)$$

where $\psi' = P_T h'_T p_J g_J + p_J g_J \sigma + P_T h'_T \sigma + p_R h'_R \sigma + \sigma^2$.

Proof. By (11), if (25) holds, the first order and second order derivative with respect to p_J are obtained as

$$\frac{\partial u_J}{\partial p_J} = \frac{P_T h'_T p_R h'_R (P_T h'_T g_J + g_J \sigma)}{\psi'^2} - y \geq 0 \quad (26)$$

$$\frac{\partial^2 u_J}{\partial p_J^2} = -\frac{P_T h'_T p_R h'_R (P_T h'_T g_J + g_J \sigma)}{\psi'^3} < 0 \quad (27)$$

It is derived that u_J achieves the maximum value when $\partial u_J / \partial p_J = 0$, and $p_J > 0$ is derived accordingly. Thus, $\arg \max_J u_J = 1$ is proved.

By (6), if (25) holds, $\forall 0 \leq p_R \leq P_R$, it is inferred that relay UAV obtains the maximum utility when $\partial u_J / \partial p_J = 0$ and

$$\begin{aligned} u(1, 1) &= \frac{P_T h'_T p_R h'_R}{\psi'} - c_1 - c_2 \\ &\geq \frac{P_T h'_T p_R h'_R}{\psi'} - c_1 = u(1, 0) \end{aligned}$$

Therefore, $\max_{\mathbf{x}} u$ with $\mathbf{x}^* = [1, 1]$ is derived. In this case, the relay UAV not only relays the maritime messages but also moves.

Remark 1. The performance bounds include four cases based on the relay UAV and the jammer jointly are provided. In Case 1, the jamming cost is high while the channel condition between ships is good, and thus both the relay UAV and the jammer do not work. In this case, the SINR, the energy consumption and the utility are expressed as (12a), (12b) and (12c). In Case 2, under the condition that the jammer is active and the relay UAV keeps silent due to high relay cost, the SINR shows as (16a), the energy consumption and the utility are calculated as (16b) and (16c). In the following two cases, it is assumed that the jammer is far from the relay UAV but close to ship B, and thus the effect of the jamming power

on the relay UAV is ignored. In Case 3, both the relay UAV and the jammer are active, the relay UAV chooses not to move due to the high moving cost as shown in (21a), the resulting SINR, energy consumption and utility are shown as (20a), (20b) and (20c). In Case 4, the UAV relays the maritime messages and moves for a better transmission link for maritime communication, the SINR, energy consumption and utility are given as indicated in (24a), (24b) and (24c).

Computational complexity: It is noted that the proposed schemes have the transmit power levels $X + 1$, moving direction levels $Z + 1$, and moving distance levels $N + 1$. Refer to [34], the computational complexity of the DQR can be approximated as $O((N + 1)(X + 1)(Z + 1)) = O(|A|)$. In the DDQNR scheme, with the batch size of the minibatch n , the number of neural nodes m in the hidden layers, and the number of action space $|A|$, the computational complexity of the DDQNR is given by $O(mn|A|)$, which can be calculated by the total number of multiplications in the forward and backward propagations. As we can see, the computational complexity of the DQR and the DDQNR increases with the number of action spaces. Moreover, the computational complexity of the DDQNR scheme also increases with the number of neural nodes and the batch size of the minibatch. The relay performance can be improved by exploiting a bigger action space. Thus, the number of action space and the DNN parameters have to be selected to balance the maritime anti-jamming communication performance and the computational complexity.

7. Performance evaluation

As shown in Fig. 3, ship A sends a message with power 100 mW to ship B aided by a UAV at 40 m height against a jammer 640 m away from Ship B. In particular, the relay UAV moves at the speed of 20 m/s [13] from the initial location (200, 300, 40) and relays the maritime messages of 1 KB with relay power chose from $\{0, 20, 40, 60, 80, 100\}$ mW to ship B against the jammer with jamming power set $\{0, 10, 20, 30, 40\}$ mW in a maritime environment with the noise power of -114 dBm [26]. According to [20], the carrier frequency is set as 5.8 GHz, the bandwidth is 20 MHz, and the antenna heights of ships A and B are set as 4.5 m. Since the center frequency follows the C-band, the path-loss exponent is set as 1.5 for the over-seawater case according to [35]. As a trade-off between the BER and the UAV energy consumption, c_1 and c_2 are set as 0.2. The DNN parameters are listed in Table 2 and simulation parameters are given by Table 3.

As the benchmark QLR as proposed in [5] applies a Q-learning to optimize the power allocation and the energy efficiency without relaying

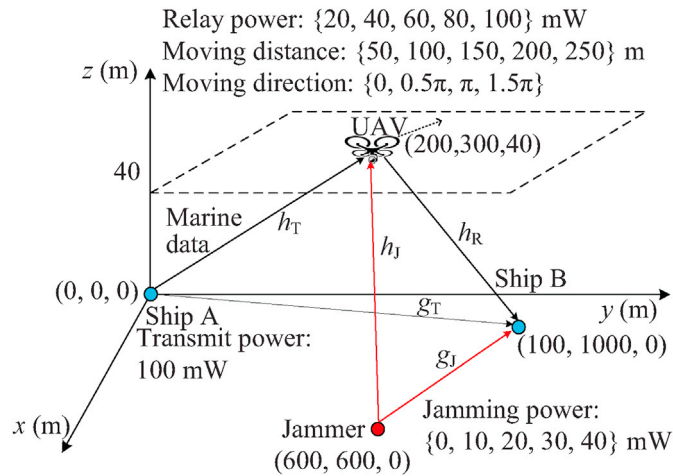


Fig. 3. Simulation settings of the maritime communication, where a UAV relays maritime messages with the power up to 100 mW against a jammer with power up to 40 mW.

Table 2

DNN parameters of DDQNR.

Layer	Input layer	Hidden layer 1	Output layer 1	Hidden layer 2	Output layer 2
Input size	4	4	256	4	256
Output size	4	256	144	256	1

Table 3

List of key simulation parameters.

Parameter	Value
Center frequency	5.8 GHz
Bandwidth	20 MHz
Noise power	-114 dBm
H_A/H_B	4.5 m
Maximum moving distance	500 m
Velocity of the UAV	20 m/s
c_1, c_2	0.2, 0.2
PL_0, η, d_0	30 dB, 1.5, 1 m
α, γ	0.2, 0.8
n, β	64, 0.001

on the channel and jamming models. As presented in Fig. 4, the DDQNR achieves the best performance by exploiting the DNN to compress the state space and dueling architecture to accelerate the learning speed. In particular, compared with the DQR and QLR at the 5000-th time slot, the DDQNR decreases the BER by 46.7% and 77.5%, reduces the energy consumption by 21.8% and 43.2%, and increases the utility by 20.8% and 76.9%, respectively.

Fig. 5 presents the performance versus the maximum jamming power over 5000 time slots. As shown in Fig. 5(c), the average utility decreases as the maximum jamming power increases. For example, the utility of the DDQNR decreases by 34.4% with the maximum jamming power ranging from 20 to 40 mW. The average BER and average energy consumption increase as the maximum jamming power increases as illustrated in Fig. 5(a) and Fig. 5(b), respectively. Because the relay UAV has to increase the relay power to ensure the BER of the received messages, which is seriously affected by the jamming power. Compared with QLR, the DQR increases the utility by 68.8% and decreases the BER and energy consumption by 63.8% and 12.6%, and the DDQNR increases the utility by 1.42 times and decreases the BER and energy consumption by 75.9% and 49.5% when the maximum jamming power is 40 mW.

As shown in Fig. 6, the average energy consumption and average utility decrease while the average BER increases as the coefficient of energy consumption increases. For example, as the coefficient of energy consumption increases from 0.1 to 0.25, the utility and energy consumption of the DDQNR decrease by 29.5% and 58.6%, and the BER increases by 5.79 times, respectively. Because the relay power decreases under high energy cost, which results in a higher BER and a lower utility. In addition, the proposed schemes achieve better performance in terms of the average utility, the average BER, and the average energy consumption over 5000 time slots under different coefficients of energy consumption compared with the QLR scheme. For example, when the coefficient of energy consumption is 0.25, the DQR increases the utility by 1.15 times and decreases the BER and energy consumption by 65.7% and 15.4%. Furthermore, the DDQNR increases the utility by 63.9% and decreases the BER and energy consumption by 20.8% and 48.1% compared with the DQR.

From Figs. 4–6, the joint optimization of both the UAV relay power and trajectory in the proposed schemes outperforms the relay power optimization in the QLR scheme. In other words, the mobility of the relay UAV affects the BER and the energy consumption performance. Because the UAV finds the optimal relay position by continually moving to increase the channel gain, and thus decrease the BER of the received signal

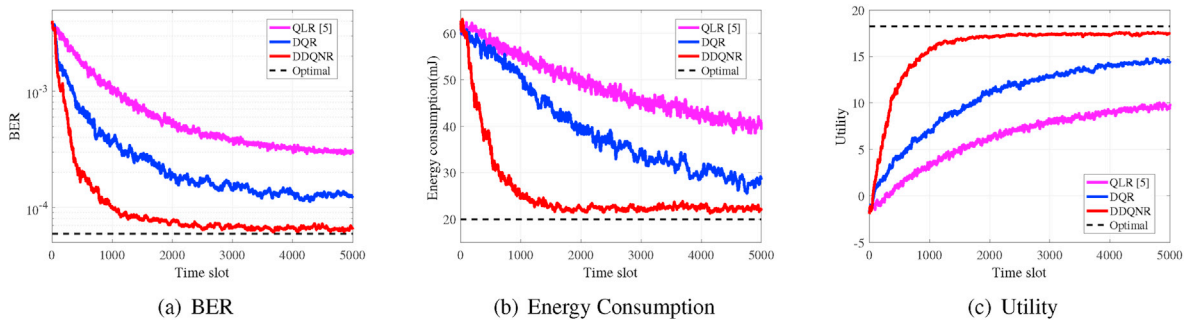


Fig. 4. Performance of the UAV aided maritime communications against jamming attacks over 5000 time slots.

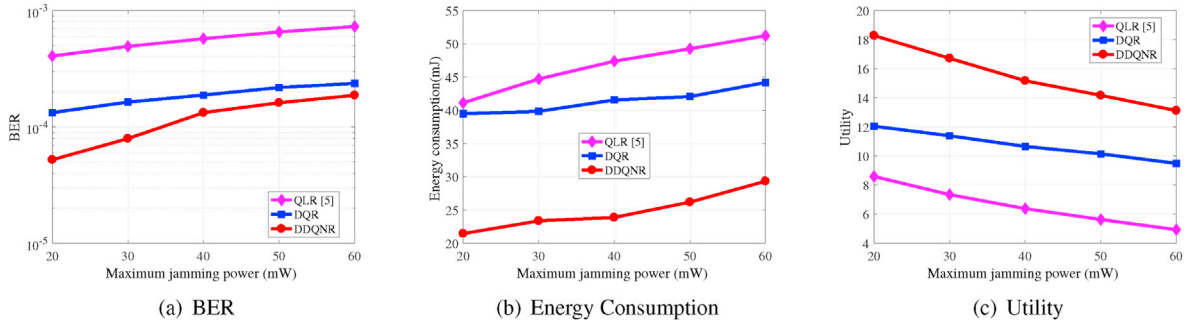


Fig. 5. Performance of the UAV aided maritime communications against jamming attacks under different maximum jamming powers.

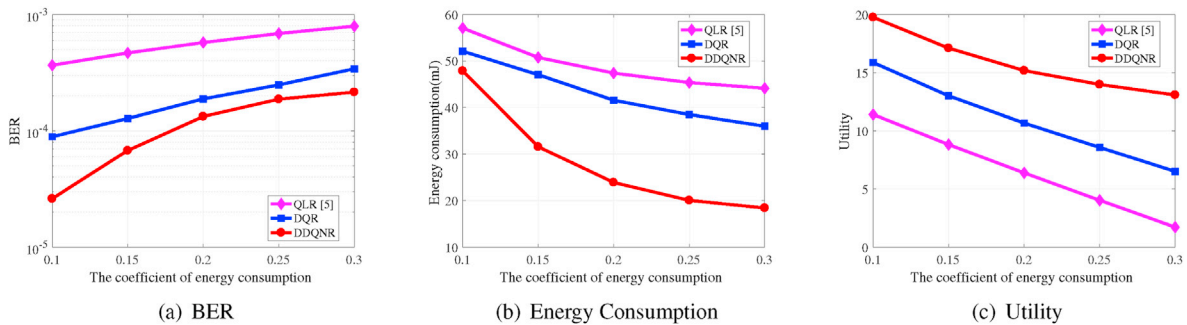


Fig. 6. Performance of the UAV aided maritime communication against jamming attacks under different coefficients of energy consumption.

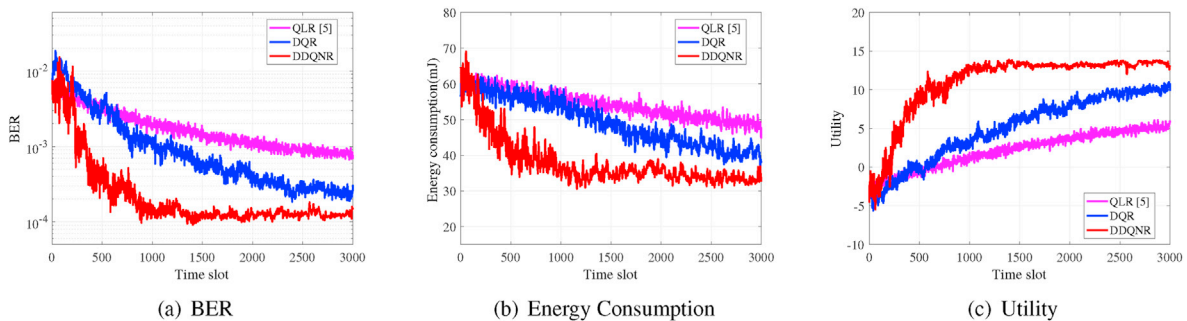


Fig. 7. Performance of the UAV aided maritime communications against jamming attacks based on the measurement data.

and the energy consumption of the relay UAV.

The channel gain is measured based on universal software radio peripherals N310 acting as the transmitter, the receiver and the jammer with the maximum signal processing bandwidth 160 MHz and frequency up to 6 GHz and maximum transmit power up to 100 mW. With the simulation results as shown in Fig. 7, the DDQNR outperforms the DQR

and QLR. In particular, compared with the QLR at the 3000-th time slot, the DQR decreases the BER by 54.4%, reduces the energy consumption by 19.6% and improves the utility by 69.9%, respectively. The DDQNR further increases the utility by 32.3%, reduces the BER by 51.6%, and saves the energy by 15.2%.

8. Conclusion

In this paper, a RL-based UAV-aided scheme have been proposed to optimize the UAV relay power and trajectory based on the BER of maritime messages, the received powers, and the UAV location for the ship-to-ship anti-jamming communication in complicated and dynamic maritime environments. The performance bounds based on the NE of the anti-jamming game and the computational complexity have been provided. Simulation results based on a UAV-aided system verify efficacy of the proposed schemes can increase utility, save energy and reduce BER compared with the benchmark QLR scheme. For example, after 5000 time slots, DQR decreases the BER by 58.1%, reduces the energy consumption by 27.5%, and improves the utility by 46.5%, and DDQNR further decreases 77.5% BER, reduces 43.2% energy consumption and increases 76.9% utility.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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